Student Performance Analytics System

Project Documentation

Python • Pandas • NumPy • CSV

1. Project Overview

The Student Performance Analytics System is a Python program that processes student marks in HTML, CSS, JavaScript, SQL, and Python. It calculates totals and averages, assigns grades, determines pass/fail results, summarizes class performance, and saves the processed report to a CSV file.

2. Objectives

- Store and read student marks using CSV files.
- Calculate each student's total and average across five subjects.
- Assign a grade based on the calculated average.
- Determine pass/fail status using the minimum mark across the five subjects.
- Summarize class performance and export the processed report.

3. Technologies Used

- **Python** - main programming language.
- **Pandas** - creates, reads, processes, and exports tabular data.
- **NumPy** - calculates averages and rounds them to two decimal places.
- **CSV** - stores input data and the generated performance report.

4. Project Files

File	Description
main.py	Python source code.
student_data.csv	Input student marks and attendance. The program creates this file from the data defined in the source code.
student_performance.csv	Output file containing marks plus Total, Average, Grade, and Result.
README.md	Project description and run instructions.

5. Grade and Result Rules

Average	Grade
90 and above	S
80 - 89.99	A
70 - 79.99	B
60 - 69.99	C
50 - 59.99	D
40 - 49.99	E

Average	Grade
Below 40	F

A student is marked **Pass** if the lowest mark across HTML, CSS, JavaScript, SQL, and Python is at least 35; otherwise, the result is **Fail**. Attendance is included in the data but is not used in the grade or pass/fail calculation.

6. Input Dataset

The dataset contains 10 students with marks (out of 100) in five subjects and an attendance percentage.

ID	Name	HTML	CSS	JS	SQL	Python	Attend.
1	Priyanka	85	78	90	76	82	92
2	Reshma	60	92	75	70	78	85
3	Jahnavi	88	65	88	97	70	98
4	Susmitha	98	85	97	99	88	75
5	Yamuna	95	96	95	87	79	94
6	JayaSree	65	98	73	78	77	88
7	Reena	86	89	55	97	98	96
8	Nithin	45	76	73	58	34	85
9	Gagan	87	54	81	97	54	74
10	Punith	83	35	39	76	45	86

7. Implementation

1. Imports Pandas and NumPy.
2. Defines student IDs, names, marks, and attendance.
3. Writes the input data to `student_data.csv` and reads it back into a DataFrame.
4. Calculates each student's total and average for the five subjects.
5. Applies the grading function and pass/fail rule.
6. Prints the detailed report and class statistics.
7. Saves the processed DataFrame to `student_performance.csv`.

8. How to Run in VS Code

1. Install Python and the Microsoft Python extension in VS Code.
2. Open the folder containing `main.py`.
3. Open **Terminal** → **New Terminal**.
4. Install the required libraries if needed:

```
python -m pip install pandas numpy
```

5. Run the program:

```
python main.py
```

6. Check the terminal for the student report and class analysis. The program writes `student_data.csv` and `student_performance.csv` in the current working folder.

9. Program Output

Student Performance Report

ID	Name	Total	Average	Grade	Result
1	Priyanka	411	82.20	A	Pass
2	Reshma	375	75.00	B	Pass
3	Jahnavi	408	81.60	A	Pass
4	Susmitha	467	93.40	S	Pass
5	Yamuna	452	90.40	S	Pass
6	JayaSree	391	78.20	B	Pass
7	Reena	425	85.00	A	Pass
8	Nithin	286	57.20	D	**Fail**
9	Gagan	373	74.60	B	Pass
10	Punith	278	55.60	D	Pass

Class Analysis

Metric	Value
Class Average	77.32
Highest Average	93.4
Lowest Average	55.6
Students Passed	9
Students Failed	1

10. Final Outcome

The program generates a structured student performance report and class-level summary. In the supplied dataset, the class average is **77.32**, the highest average is **93.40** (Susmitha), the lowest average is **55.60** (Punith), and **9 students pass while 1 student fails** under the program's current rule.

11. Note

The pass/fail decision follows the code exactly: it checks whether the minimum subject mark is at least 35. Therefore, a student can have an average below 60 and still be marked Pass if every subject mark is 35 or higher.

Appendix: Source Code (main.py)

```
import pandas as pd
import numpy as np

data = {
    "Student_ID": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "Name": [
        "Priyanka", "Reshma", "Jahnavi", "Susmitha", "Yamuna", "JayaSree","Reena", "Nithin", "Gagan", "Punith"
    ],
    "HTML"      : [85,60,88,98,95,65,86,45,87,83],
    "CSS"       : [78,92,65,85,96,98,89,76,54,35],
    "JAVASCRIPT": [90,75,88,97,95,73,55,73,81,39],
    "SQL"       : [76,70,97,99,87,78,97,58,97,76],
    "PYTHON"    : [82,78,70,88,79,77,98,34,54,45],
    "Attendance": [92,85,98,75,94,88,96,85,74,86]
}

pd.DataFrame(data).to_csv("student_data.csv", index=False)
df = pd.read_csv("student_data.csv")

print("\nSTUDENT PERFORMANCE ANALYTICS SYSTEM")
print("\nSTUDENT DATA")
print(df.to_string(index=False))

subjects = ["HTML", "CSS", "JAVASCRIPT", "SQL", "PYTHON"]
df["Total"] = df[subjects].sum(axis=1)
df["Average"] = np.round(df[subjects].mean(axis=1), 2)
def calculate_grade(avg):
    if avg >= 90:
        return "S"
    elif avg >= 80:
        return "A"
    elif avg >= 70:
        return "B"
    elif avg >= 60:
        return "C"
    elif avg >= 50:
        return "D"
    elif avg >= 40:
        return "E"
    else:
        return "F"

df["Grade"] = df["Average"].apply(calculate_grade)
df["Result"] = df[subjects].min(axis=1).apply(
    lambda mark: "Pass" if mark >= 35 else "Fail"
)
print("\nSTUDENT PERFORMANCE REPORT")
print(df[[
    "Student_ID", "Name", "Total",
    "Average", "Grade", "Result"
]].to_string(index=False))
print("\nCLASS ANALYSIS")
print("Class Average:", round(df["Average"].mean(), 2))
print("Highest Average:", df["Average"].max())
print("Lowest Average:", df["Average"].min())
print("Students Passed:", (df["Result"] == "Pass").sum())
print("Students Failed:", (df["Result"] == "Fail").sum())
df.to_csv("student_performance.csv", index=False)

print("\nCSV files created successfully......")
```